DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**CENTRAL LINE MONITORING, CHANGING FLUIDS, TUBING AND DRESSING**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-NICU-014 (v2)**		
Last Version Approval Date	**28/1/2022**	Number of Pages	**5**

	NAME	POSITION	SIGNATURE	DATE
Prepared by:	**MS. JOICY JOY**	NICU Quality Representative		27/2/2024
	MS. AISHA ZAIBI SAAD AL-SHUMRANI	Neonatal Intensive Care Unit, Head Nurse		27/2/2024
	DR. TAYSEER HASSAN	Neonatology Resident		27/02/2024
Reviewed by:	**DRA. FAWZIA MOHAMMED FARAHAT**	Head, Neonatology and Pediatric Department		14/03/24
	MR. MANSOUR AHMED AL-ZAHRANI	Head, Quality Management and Patient Safety		14.3.24
Approved by:	**MR. SULTAN ALI AL-GAMDI**	Nursing Director		19-3-24
	DR. WAEL KAMEL ARBA	HED Assistant for Medical Affairs		19.3.24

APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
19/3 / 2024	7 / 4 / 2024	TRIENNIAL	18 / 3 / 2027

Baha_Cluster
Healthcluster-baha@moh.gov.sa
مستشفى المخواة العام
Al-Mikhwah General Hospital

1.0 PURPOSE

1.1. To provide an aseptic, systemic approach in changing a central line dressing to prevent infection at the Intravenous (IV) site and the introduction of microorganisms into the blood stream.

2.0 DEFINITION/ABBREVIATION

2.1. **Central venous catheter (CVC)** or a central line is a long, thin, flexible tube used to give medicines, fluids, nutrients, or blood products over a long period of time, usually several weeks or more. A catheter is often inserted in the arm or chest through the skin into a large vein. The catheter is threaded through this vein until it reaches a large vein near the heart.

2.2. **Types of Central Venous Catheters**:

2.2.1. **Tunneled Catheters** (commonly used are Hickman, Broviac and Groshong catheters) – is inserted into a central vein (usually the subclavian, then the superior vena cava) and subcutaneously tunneled to an exit site approximately 4 inches (10 cm) from the insertion site. The exit site is typically located in the chest, making the access port less visible than if they were directly protruding in the neck. Passing the catheter under the skin helps to prevent infection & provides stability.

2.2.2. **Non-Tunneled catheters** (usually used are Quinton catheters) – commonly called percutaneous, may be inserted in femoral, jugular, or subclavian veins with 1 to 4 lumens, dwell time usually less than 1 month.

2.2.3. **Implanted Port** – is similar to a tunneled catheter but is left entirely under the skin. Medicines are injected through the skin into the catheter. Surgically implanted infusion. Ports are placed below the clavicle, with the catheter threaded into the right atrium through large vein. Once implanted, the port is accessed via non – coring "Huber" needles inserted through the skin. Ports are easy to maintain (compared to CVC or PICC) and typically used on patients requiring occasional venous access over a long duration course of therapy.

2.2.4. **PICC line** (Peripherally Inserted Central Catheter) – is a central venous catheter inserted in a peripheral vein, such as the cephalic vein, basilic vein, or brachial vein and then advanced through increasingly larger veins, toward the heart until the tip rests in the distal superior vena cava or cavoatrial junction.

3.0 RESPONSIBILITY

3.1. Staff Nurses

3.2. Physician

4.0 POLICY

4.1. Central venous catheters allow long term administration in situations requiring safe, repeated access to the venous system. Multiple blood samples can be drawn through it without repeated venipuncture. However, CV therapy also increases the risk of complications, such as pneumothorax, sepsis, thrombus formation, vessel & adjacent organ perforation. Sterile technique is highly important in managing CVC line.

4.2. To prevent infection, stringent cleaning of the catheter insertion site is a must using povidone-iodine, but chlorhexidine if available is more effective.

4.3. Topical antibiotic ointment or cream should not be used on the insertion site due to the potential of promoting a fungal infection or antimicrobial resistance.

4.4. If multi-lumen catheter is used, a catheter port must be selected and marked for TPN only and other port must be labeled.

4.5. Strict sterile technique must be observed in changing intravenous fluids & tubing in central line.

4.6. IV tubing connected to central line must be changed every 24 hours.

4.7. Central line dressing should be changed every 48 hours.

5.0 PROCEDURE (S)

5.1. General:

5.1.1. Check the correct identity of the neonate.

5.1.2. Double check by two nurses the IV fluids against the physician's order.

5.1.3. Perform hand hygiene and organize equipment.

5.1.4. Procedure should be done in strict sterile technique.

5.1.5. Wear masks and head cover.

5.2. **Monitoring & Performing Maintenance**

5.2.1. Wash hands thoroughly.

5.2.2. Label each lumen of multi lumen catheter with name of fluid / medication infusing.

5.2.3. Flush lumens without continuous fluid infusions and capped every 8 hours with heparin solution (usually 1:100 dilution) or normal solution.

5.2.3.1. Depending on length of tubing and size of catheter, use 1 to 3 mL of flush solution.

5.2.3.2. Use 6 mL or ordered amount of flush for Hickman catheter & short small needle (25 gauge)

5.2.3.3. For PICC lines, use a 10-cc syringe or larger for flushing.

5.2.4. Flush tubing between infusion of medications and drawing of blood, first using saline and then Heparinized solution.

5.2.5. Always aspirate before infusing medications or flushing

5.2.6. Monitor for clot formation in lumen. If resistance is met when flushing tubing, do not force.

Baha_Cluster
Healthcluster-baha@moh.gov.sa
مستشفى المخواة العام
Al-Mikhwah General Hospital

5.2.7. Aspirate and remove clot if possible; if not, notify physician.

5.2.8. Monitor respirations and breath sounds every 4 hours.

5.2.9. Maintain IV fluids above heart level. Do not allow fluid to run out, and air to enter tubing.

5.3. **Tubing Change:**

5.3.1. Prepare fluid and tubing aseptically.

5.3.2. Don mask & sterile gloves.

5.3.3. Expose catheter hub or rubber port of multi lumen catheter.

5.3.4. For centrally inserted lines:

5.3.4.1. Ask patient to gently turn head to opposite side, take a deep breath and bear down (Valsalva's maneuver).

5.3.4.2. Disconnect old tubing and quickly connect new one.

5.3.4.3. Open fluid and adjust to appropriate infusion rate.

5.3.4.4. Remove gloves, discard equipment, and position patient comfortably.

5.4. **Dressing Change:**

5.4.1. Explain procedure to patient.

5.4.2. Perform hand hygiene and gather equipment.

5.4.3. Prepare equipment/supplies on sterile field.

5.5. **Changing Fluids:**

5.5.1. Open the IV tubing system in sterile technique and place on the sterile field.

5.5.2. Connect a triple connector to the IV tubing if more than one IV solution is infusing through the site. Extra triple ports must be flushed with normal saline, clamped and capped with luer-lock caps to prevent air pockets or bubbles from forming during fluid priming.

5.5.3. Under sterile conditions spike the IV fluids, fill the buretrol with fluid by gently squeezing the buretrol and drip chamber, & purging the IV fluid through the tubing.

5.5.4. Disconnect the old IV tubing from the positive-flow cap on the central catheter, then clean the positive flow cap with alcohol for 10 seconds, allowing the alcohol to dry before connecting the new IV tubing to the central catheter.

5.5.5. At this point, the sterile procedure is complete. The nurse will attach the tubing to the IV administration pump and set the rate & volume of fluid to be infused. Clear the previous volume infused on the pump prior to administration if a new type of fluid is being hung.

Baha_Cluster
Healthcluster-baha@moh.gov.sa
مستشفى المخواة العام
Al-Mikhwah General Hospital

5.5.6. Medications to be administered via a central venous line will be given via microbore tubing with a needleless cap. A needleless cap should be placed between the triple connector and the medication line, and also at the end of the medication line where the syringe will be attached.

5.5.7. When changing IV fluid only, don a head cover, mask and clean gloves. Clean around the spike with alcohol for 10 seconds, allow to dry before disconnecting the syringe or the IV bag. Spike the new bag or attach the syringe.

5.5.8. All solutions infused via central lines must be infused via a neonatal IV pump. Central line IV tubing will be changed every 24 hours.

5.5.9. Blood / blood products may be given through the central line with a physician's order.

5.6. **Documentation**:

5.6.1. Record the condition of the site and dressing when changing the IV fluids.

5.6.2. Place initials, date and time of the IV fluid being hung and label.

5.6.3. If hanging TPN, place the pharmacy label on the back of the original TPN order.

5.6.4. Date and initial the IV fluid change on the medication sheet.

5.6.5. Label triple connectors, IV tubing and medication pumps with the name of the IV fluids

6.0 EQUIPMENTS

6.1. Infusion Set
6.2. 3-way Stopcock
6.3. Fosiflush
6.4. Alcohol swab

7.0 ATTACHMENT/FORMS

7.1. Nurses Progress Notes (GDOH-NUR-NPN-203)
7.2. Medication Administration Record (GDOH-NUR-MAR-191)

8.0 REFERENCES

8.1. Lippincott Manual of Nursing Practice, 9th edition, 2010, by Sandra Nettina

8.2. Taylor's Clinical Nursing Practice, by Pamela Evans-Smith

8.3. http://en.wikepedia.org/wike/central_venous_catheter

9.0 REVISION CLAUSE

9.1. Modification of policy format, lay out and policy coding

Baha_Cluster
Healthcluster-baha@moh.gov.sa
مستشفى المخواة العام
Al-Mikhwah General Hospital

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

0 000000 002035

SN

KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: |_|_|_|_|_|_|_|_|_|_| رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

MEDICATION ADMINISTRATION RECORD

Admission date:	Weight:	Height:

ALLERGIES: ❒ NKA ❒ Yes (If yes, include allergy to medications, food & blood products):

Nurse's Notes:

1. When a new prescription sheet is used, all current treatments must be fully re-written, re-signed.
2. Check for entries in every section in order to avoid omissions.
3. Follow medication administration and transcription order policy
4. In the event of a medicine not given, enter the appropriate code in the administration box available in page 2

ONCE ONLY PRESCRIPTION (STAT)

Date	Time Ordered	Drug	Dose	Route	Time given	Assigned Nurse Initial	Staff Nurse Initial

INTRAVENOUS SOLUTION

Date	Time Ordered	Types/ Volume of Fluids	Types/ Amount of Additives	Rate	Starting time	Assigned Nurse Initial	Staff Nurse Initial	Date/Time Discontinued

NOTE: Utilize the last page if the above spaces provided are insufficient.

NURSES IDENTIFICATION:

Name	Job Number	Initial	Name	Job Number	Initial	Name	Job Number	Initial

0 000000 001915

SN

Name:_____ الاسم:	MRN: رقم الملف الطبي:

NON- ADMINISTRATION
If a dose is not administered by a nurse for any reason, the nurse should initial the administration record and enter the appropriate code number.

CODES:
1 = Allergic Reaction
2 = Patient Fasting
3 = Omitted for Clinical Reasons
4 = Patient Refused
5 = Patient Unavailability
6 = Drug Unavailability
7 = Self Administration

Medication Name:

Dose	Route	Frequency
Initial dose :	Starting date:	
RN Name:	RN Initial:	
COUNTERSIGNED BY: (STAFF NURSE)		
Name:	Initial:	
Time : Date:	Other instructions:	
Time drug discontinued: Date: / /		

DATE TIME							

Medication Name:

Dose	Route	Frequency
Initial dose:	Starting date:	
RN Name:	RN Initial:	
COUNTERSIGNED BY: (STAFF NURSE)		
Name:	Initial:	
Time : Date:	Other instructions:	
Time drug discontinued: Date: / /		

DATE TIME							

Medication Name:

Dose	Route	Frequency
Initial dose:	Starting date:	
RN Name:	RN Initial:	
COUNTERSIGNED BY: (STAFF NURSE)		
Name:	Initial:	
Time : Date:	Other instructions:	
Time drug discontinued: Date: / /		

DATE TIME							

Medication Name:

Dose	Route	Frequency
Initial dose:	Starting date:	
RN Name:	RN Initial:	
COUNTERSIGNED BY: (STAFF NURSE)		
Name:	Initial:	
Time : Date:	Other instructions:	
Time drug discontinued: Date: / /		

DATE TIME							

Medication Name:

Dose	Route	Frequency
Initial dose:	Starting date:	
RN Name:	RN Initial:	
COUNTERSIGNED BY: (STAFF NURSE)		
Name:	Initial:	
Time : Date:	Other instructions:	
Time drug discontinued: Date: / /		

DATE TIME							

Name: _____ :الاسم	MRN: _____ رقم الملف الطبي:

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Medication Name:	**DATE TIME**							
Dose / Route / Frequency								
Initial dose: / Starting date:								
RN Name: / RN Initial:								
COUNTERSIGNED BY: (STAFF NURSE)								
Name: / Initial:								
Time : Date: / Other instructions:								
Time drug discontinued: Date: / /								

Name:_____ الاسم

MRN: |_|_|_|_|_|_|_|_|_|_| رقم الملف الطبي:

HIGH ALERT MEDICATION

Medication Name:

Dose	Route	Frequency
Initial dose:	Starting date:	
RN Name:		RN Initial:
DOUBLE CHECKED BY: (STAFF NURSE)		
Name:		Initial:
Time :	Date:	Other instructions:
Time drug discontinued: Date: / /		

DATE TIME							

Medication Name:

Dose	Route	Frequency
Initial dose:	Starting date:	
RN Name:		RN Initial:
DOUBLE CHECKED BY: (STAFF NURSE)		
Name:		Initial:
Time :	Date:	Other instructions:
Time drug discontinued: Date: / /		

DATE TIME							

Medication Name:

Dose	Route	Frequency
Initial dose:	Starting date:	
RN Name:		RN Initial:
DOUBLE CHECKED BY: (STAFF NURSE)		
Name:		Initial:
Time :	Date:	Other instructions:
Time drug discontinued: Date: / /		

DATE TIME							

Medication Name:

Dose	Route	Frequency
Initial dose:	Starting date:	
RN Name:		RN Initial:
DOUBLE CHECKED BY: (STAFF NURSE)		
Name:		Initial:
Time :	Date:	Other instructions:
Time drug discontinued: Date: / /		

DATE TIME							

Medication Name:

Dose	Route	Frequency
Initial dose:	Starting date:	
RN Name:		RN Initial:
DOUBLE CHECKED BY: (STAFF NURSE)		
Name:		Initial:
Time :	Date:	Other instructions:
Time drug discontinued: Date: / /		

DATE TIME							

0 000000 001915 SN

Name: _____ الاسم:	MRN: رقم الملف الطبي:

INTRAVENOUS SOLUTION

Date	Time Ordered	Types/ Volume of Fluids	Types/ Amount of Additives	Rate	Starting time	Assigned Nurse Initial	Staff Nurse Initial	Date/Time